fn new_negative_binomial_rdp_curve

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Proves soundness of `fn new_negative_binomial_rdp_curve` in `mod.rs`.

1 Hoare Triple

Precondition

`base_curve` upper-bounds the base Rényi divergence at every order > 1, `eta` ≥ 0, `mean` > 0 is finite, and `log_inv_gamma_hi_from_x(x)` succeeds whenever it is called.

Pseudocode

```
def new_negative_binomial_rdp_curve(base_curve, eta: float, x: float, mean: float):
    def curve(alpha: float):
        if alpha <= 1.0:  #
            return float("inf")
        eps_alpha = base_curve(alpha)  # propagates helper errors
        one_plus_eta_hi = 1.0.inf_add(eta)  # propagates helper errors
        alpha_minus_one_lo = alpha.neg_inf_sub(1.0)  # propagates helper errors
        inv_alpha_lo = 1.0.neg_inf_div(alpha)  # propagates helper errors
        one_minus_inv_alpha_hi = 1.0.inf_sub(inv_alpha_lo)  # propagates helper errors
        coeff1_hi = one_plus_eta_hi.inf_mul(one_minus_inv_alpha_hi)  # propagates helper errors
        term1_hi = coeff1_hi.inf_mul(eps_alpha)  # propagates helper errors
        log_inv_gamma_hi = log_inv_gamma_hi_from_x(x)  # propagates helper errors
        term2_num_hi = one_plus_eta_hi.inf_mul(log_inv_gamma_hi)  # propagates helper errors
        term2_hi = term2_num_hi.inf_div(alpha)  # propagates helper errors
        log_mean_hi_ = log_mean_hi(mean)  # propagates helper errors
        term3_hi = log_mean_hi_.inf_div(alpha_minus_one_lo)  # propagates helper errors
        return eps_alpha.inf_add(term1_hi).inf_add(term2_hi).inf_add(term3_hi)  # propagates helper errors
    return curve
```

Postcondition

Theorem 1.1. For every $\alpha > 1$, the returned function either returns `Err(e)` due to a propagated helper error, or returns an upper bound on the implemented Theorem 2 expression

$$\epsilon(\alpha) \;+\; (1+\eta)\left(1-\frac{1}{\alpha}\right)\epsilon(\alpha) \;+\; \frac{(1+\eta)\log(1/\gamma)}{\alpha} \;+\; \frac{\log(\mathbb{E}[K])}{\alpha-1},$$

where $\epsilon(\cdot)$ is the base Rényi divergence curve, $\gamma = 1 - e^{-x}$, and the input `mean` upper-bounds $\mathbb{E}[K]$.

2 Proof

Fix $\alpha > 1$.

Lemma 2.1. If $\alpha \leq 1$, the returned value is $+\infty$, which is a valid upper bound.

Proof. The expression is finite or infinite, and $+\infty$ upper-bounds either case. □

Lemma 2.2. If $\alpha > 1$ and the function returns `Ok(out)`, then `out` upper-bounds the displayed expression.

Proof. Let $\epsilon(\cdot)$ denote the exact base Rényi divergence curve. By precondition,

$$\texttt{eps_alpha} \geq \epsilon(\alpha).$$

Also $\texttt{one_plus_eta_hi} \geq 1 + \eta$, $\texttt{inv_alpha_lo} \leq 1/\alpha$, and hence

$$\texttt{one_minus_inv_alpha_hi} \geq 1 - \frac{1}{\alpha}.$$

All of these quantities are nonnegative, so upward-rounded multiplication yields

$$\texttt{term1_hi} \geq (1+\eta)\left(1 - \frac{1}{\alpha}\right)\epsilon(\alpha).$$

By the proof of `log_inv_gamma_hi_from_x`,

$$\texttt{log_inv_gamma_hi} \geq \log(1/\gamma).$$

Since `alpha` is a positive exact denominator,

$$\texttt{term2_hi} \geq \frac{(1+\eta)\log(1/\gamma)}{\alpha}.$$

By the proof of `log_mean_hi`,

$$\texttt{log_mean_hi_} \geq \log(\texttt{mean}) \geq \log(\mathbb{E}[K]).$$

Also $\texttt{alpha_minus_one_lo} \leq \alpha - 1$ with a positive denominator, so

$$\texttt{term3_hi} \geq \frac{\log(\mathbb{E}[K])}{\alpha - 1}.$$

Finally, `out` is formed by upward-rounded addition of `eps_alpha`, `term1_hi`, `term2_hi`, and `term3_hi`, so it upper-bounds the full displayed expression. □

Proof of Theorem 1.1. By Lemma 2.1 for $\alpha \leq 1$, and Lemma 2.2 otherwise. □